

GradeAnalyzer

```
1  import java.io.*;
2  import java.util.ArrayList;
3
4  public class GradeAnalyzer {
5
6      public static void main(String[] args) {
7          // Step 1: read scores from file
8          ArrayList<Integer> scores = readScores("scores.txt");
9
10         // Step 2: calculate statistics
11         double classAvg = calculateAverage(scores);
12         int minScore = Integer.MAX_VALUE;
13         int maxScore = Integer.MIN_VALUE;
14         for (int score : scores){
15             if(minScore > score){
16                 minScore = score;
17             } if (maxScore < score){
18                 maxScore = score;
19             }
20         }
21         int countA, countB, countC, countD, countF;
22         countA = countB = countC = countD = countF = 0;
23         for (int score : scores){
24             if(score < 60){
25                 countF += 1;
26             } else if (score < 70){
27                 countD++;
28             } else if (score < 80){
29                 countC++;
30             } else if (score < 90){
31                 countB++;
32             } else {
33                 countA++;
34             }
35         }
36         // Step 3: write and print report
37         writeReport(scores, classAvg, maxScore, minScore,
38                     countA, countB, countC, countD, countF, "report.txt");
39     }
40
41     // Returns a list of valid scores read from the file
42     public static ArrayList<Integer> readScores(String filename) {
43         ArrayList<Integer> scores = new ArrayList<>();
44         try(BufferedReader reader = new BufferedReader(new
45             FileReader(filename))){
46             String line;
47             while((line = reader.readLine()) != null){
48                 line = line.trim();
49                 if(line.isEmpty()){
50                     continue;
51                 }
52                 scores.add(Integer.parseInt(line));
53             }
54         } catch (IOException e) {
55             e.printStackTrace();
56         }
57         return scores;
58     }
59
60     // Calculate the average of a list of scores
61     public static double calculateAverage(ArrayList<Integer> scores) {
62         if(scores.isEmpty()) return 0;
63         double sum = 0;
64         for (int score : scores) {
65             sum += score;
66         }
67         return sum / scores.size();
68     }
69
70     // Write a report to a file
71     public static void writeReport(ArrayList<Integer> scores, double classAvg,
72                                    int maxScore, int minScore, int countA,
73                                    int countB, int countC, int countD, int countF,
74                                    String filename) {
75         try {
76             FileWriter writer = new FileWriter(filename);
77             writer.write("Grade Report\n");
78             writer.write("Total Scores: " + scores.size() + "\n");
79             writer.write("Average Score: " + classAvg + "\n");
80             writer.write("Maximum Score: " + maxScore + "\n");
81             writer.write("Minimum Score: " + minScore + "\n");
82             writer.write("Grade Distribution:\n");
83             writer.write("A: " + countA + ", B: " + countB + ", C: " + countC + ", D: " + countD + ", F: " + countF + "\n");
84             writer.close();
85         } catch (IOException e) {
86             e.printStackTrace();
87         }
88     }
89 }
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50         }
51         try {
52             scores.add(Integer.parseInt(line));
53             //System.out.println("Added score " +
Integer.parseInt(line));
54         } catch (NumberFormatException e) {
55             System.out.println("Warning: skipping invalid score: " +
line);
56         }
57     }
58     } catch (IOException e) {
59         System.out.println("Failed to read file :(");
60     }
61     return scores;
62 }
63
64 // Returns the average of a list of scores, or 0.0 if the list is empty
65 public static double calculateAverage(ArrayList<Integer> scores) {
66     int scoreCount = scores.size();
67     double avgScore = 0.0;
68     for(int score : scores){
69         avgScore += score;
70     }
71     //System.out.println("Avg Score = " + (avgScore / scoreCount));
72     return scoreCount == 0 ? 0.0 : avgScore / scoreCount;
73 }
74
75 // Writes and prints the report
76 public static void writeReport(ArrayList<Integer> scores,
77                                double avg, int high, int low,
78                                int countA, int countB, int countC,
79                                int countD, int countF,
80                                String outputFile) {
81     String report = String.format(
82         "=== Grade Analysis Report ===\n"
83         + "Total scores processed:  %d\n"
84         + "Average score: %.2f\n"
85         + "Highest score: %d\n"
86         + "Lowest score: %d\n"
87         + "%nGrade distribution:%n"
88         + "A (90-100):  %d\n"
89         + "B (80-89):   %d\n"
90         + "C (70-79):   %d\n"
91         + "D (60-69):   %d\n"
92         + "F (below 60): %d\n",
93         scores.size(), avg, high, low,
94         countA, countB, countC, countD, countF);
95     System.out.print(report);
96
97     try(BufferedWriter writer = new BufferedWriter(new
FileWriter(outputFile))){
98         writer.write(report);
99     }catch(IOException e){
100         System.out.println("Error in writing to report");
```

101		}
102	}	
103	}	